6 A potential divider is made by connecting a light-dependent resistor (LDR) and a thermistor in series. Fig. 6.1 shows the potential divider, a voltmeter and a direct current (d.c.) power supply connected into a circuit.

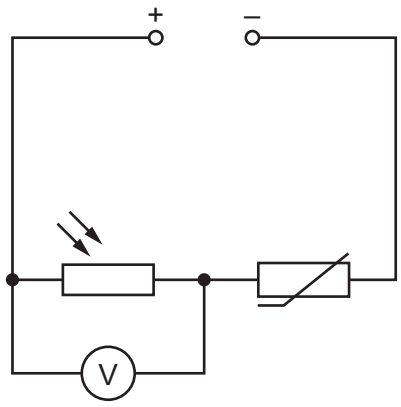

Fig. 6.1

The voltmeter measures the potential difference (p.d.) across the LDR.

(a) Define potential difference (p.d.).

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..... [2]

(b) The electromotive force (e.m.f.) of the supply is *E*.

Describe how the p.d. across the thermistor can be determined using the reading on the voltmeter.

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..... [1]

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(c) The resistance of the LDR decreases and the resistance of the thermistor increases.

(i) State what has happened to the light intensity incident on the LDR and the temperature of the thermistor.

intensity of incident light on LDR: .....

temperature of thermistor: .....
[1]

(ii) Explain what happens to the reading on the voltmeter.

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..... [3]

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